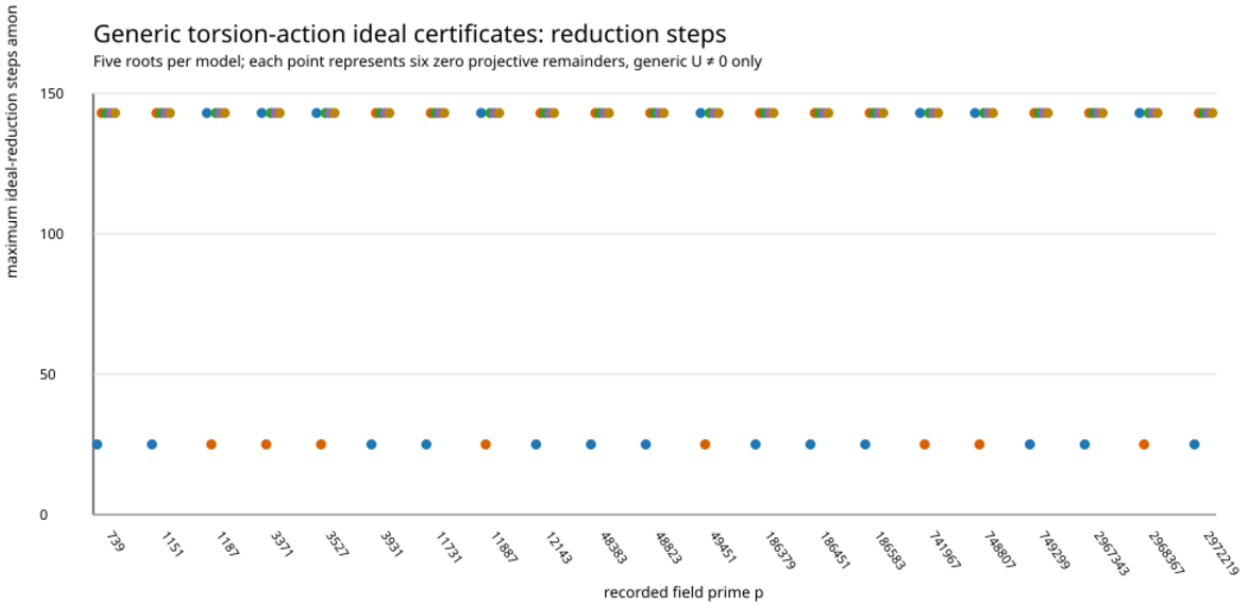
N16 generic two-torsion action: exact polynomial-ideal certificates

Frozen 21-model Rosenhain packet

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The native producer and separate native verifier checked all 105 recorded model/root pairs. On the degree-two locus $U = u(r) \neq 0$, all 630 cleared projective cross-products reduce to zero modulo the two Mumford equations. The retained coefficient field, matrices, generators, cross-products, quotient witnesses, and zero remainders are in `results/generic-ideal.json`. This is a generic-branch certificate, not an all-state law or a new collector measurement.

Quantity	Verified value
Recorded prime-field models	21
Rational two-torsion roots per model	5
Generic model/root pairs	105
Projective cross-products	630
Nonzero polynomial remainders	0
Quotient-division residual remainders	0 of 315
Clearing denominator	U^6 on every pair
Maximum sparse cross-product terms	288
Maximum reduction steps per cross-product	143
Exceptional $U = 0$ branch theorem	null
Degree-one/zero geometric action theorem	null
New collector/rho cost ratio	null



Write $u = x^2 + a_1x + a_0$, $v = b_1x + b_0$, and $f \bmod u = R_0 + R_1x$. The ideal generators are

$$E_0 = b_0^2 - a_0b_1^2 - R_0, \qquad E_1 = 2b_0b_1 - a_1b_1^2 - R_1.$$

For each root r, put $U = u(r)$, $V = v(r)$, $L = (Uv - Vu)/U$. Polynomial division of $U^2f - (Uv - Vu)^2$ by $u(x - r)$ yields $q_2x^2 + q_1x + q_0$ with $q_2 = U^2$ modulo (E_0, E_1). Thus the generic translated divisor has $u' = x^2 + (q_1/U^2)x + q_0/U^2$.

Its second Mumford coefficient comes from $-L \bmod u'$. The cancelled Kummer chart is converted by the recorded nonsingular matrix C; the candidate action matrix is M_r.

The producer clears U^6 in the translated four-vector K' and constructs six polynomials $F_{ij} = K'_i(M_rK)_j - K'_j(M_rK)_i$ for $0 \le i < j < 4$. Every retained certificate explicitly supplies A_{ij}, B_{ij} such that $F_{ij} = A_{ij}E_0 + B_{ij}E_1$ in $\mathbb{F}_p[a_0, a_1, b_0, b_1]$. A separate Rust checker reconstructs f, C, M_r, the translated numerator, all six F_{ij}, and these polynomial equalities from the frozen inputs. In all 105 rows, the first transformed coordinate is $W_0 = \kappa_r U$ with $\kappa_r \neq 0$; hence the projective relation is defined on $U \neq 0$.

The finite exceptional replay in the main N16 report remains separate evidence. This certificate does not settle $U = 0$, degree-one or zero divisor charts, their geometric action identities, relation-label preservation by a complete collector, or its cold cost against matched rho. The standardized 105-row CSV, review graph, full raw certificate, exact command, and SHA-256 list accompany this PDF.